

STAT 220B — Midterm (Makeup)

Time: 2 hours Open notes

This exam is graded out of 115 points; final scores will be capped at 100. You may use the statement of any earlier part of a question in later parts, even if you did not complete the earlier part.

1. (30 points) Risk sets, admissibility, and randomization.

Let $\Theta = \{\theta_0, \theta_1\}$ and $\mathcal{A} = \{a_0, a_1\}$. The sample space is $\mathcal{X} = \{1, 2, 3\}$. The sampling distributions are

$$\mathbb{P}_{\theta_0}(X = 1) = \frac{1}{2}, \quad \mathbb{P}_{\theta_0}(X = 2) = \frac{3}{10}, \quad \mathbb{P}_{\theta_0}(X = 3) = \frac{1}{5},$$

$$\mathbb{P}_{\theta_1}(X = 1) = \frac{1}{10}, \quad \mathbb{P}_{\theta_1}(X = 2) = \frac{1}{5}, \quad \mathbb{P}_{\theta_1}(X = 3) = \frac{7}{10}.$$

The loss matrix is

	a_0	a_1
θ_0	0	2
θ_1	4	0

The 8 non-randomized decision rules are labeled δ_{abc} , where $a, b, c \in \{0, 1\}$ are the actions taken at $X = 1, 2, 3$ respectively.

- (a) Compute the risk vector $(R(\theta_0, \delta_{abc}), R(\theta_1, \delta_{abc}))$ for each of the 8 rules, as exact fractions. (6 points)
- (b) Identify which of the 8 rules are admissible. For each inadmissible rule, exhibit a dominating rule. (5 points)
- (c) For each of the following rules, determine the set of priors $\pi(\theta_0) = \pi_0 \in [0, 1]$ for which the rule is Bayes:

$$\delta_{011}, \quad \delta_{001}, \quad \delta_{010}.$$

Give your answers as exact intervals (or as the empty set). (12 points)

- (d) Use part (c) to characterize, with brief justification, the set of non-randomized rules that are Bayes for at least one prior $\pi_0 \in [0, 1]$. Compare this set with your answer to (b). (4 points)
- (e) Consider the randomized rule $\delta^* = \lambda\delta_{001} + (1 - \lambda)\delta_{011}$. Find the unique $\lambda^* \in (0, 1)$ for which $R(\theta_0, \delta^*) = R(\theta_1, \delta^*)$, and state the common risk value. (3 points)

2. (25 points) Utility theory and risk aversion.

A decision maker has utility function $U(w) = \log w$ for wealth $w > 0$ (natural logarithm). The decision maker's current wealth is $w_0 = 100$.

A friend offers the following gamble: a fair coin is tossed; if heads, the decision maker loses 50; if tails, the decision maker gains 60. Let W be the decision maker's wealth after the gamble.

- (a) Compute $\mathbb{E}[W]$ and $\mathbb{E}[U(W)]$. (5 points)
- (b) Should the decision maker accept the gamble? Justify. (5 points)
- (c) Compute the certainty equivalent CE of W , defined by $U(\text{CE}) = \mathbb{E}[U(W)]$, in closed form. (6 points)
- (d) Compute the risk premium $\text{RP} = \mathbb{E}[W] - \text{CE}$ in closed form. (4 points)
- (e) Explain why $\text{RP} > 0$ for this decision maker. (5 points)

3. (30 points) Rao–Blackwell, strict convexity, and admissibility.

Let $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(\theta)$ with $\theta \in (0, 1)$ and $n \geq 2$. We wish to estimate $g(\theta) = \theta(1 - \theta) = \text{Var}(Y_1)$ under squared-error loss. Let $T = \sum_{i=1}^n Y_i$.

- (a) State the distribution of T and verify, via the factorization criterion, that T is sufficient for θ . (4 points)

(b) Show that

$$\delta_0(Y_1, Y_2) = \frac{(Y_1 - Y_2)^2}{2}$$

is unbiased for $g(\theta)$. (5 points)

(c) Compute the Rao–Blackwellized estimator

$$\delta_1(T) = \mathbb{E}[\delta_0(Y_1, Y_2) \mid T].$$

(10 points)

(d) Show that $\text{Var}(\delta_0 \mid T = t) > 0$ for every $t \in \{1, 2, \dots, n-1\}$, and conclude that $R(\theta, \delta_1) < R(\theta, \delta_0)$ strictly for every $\theta \in (0, 1)$. (8 points)

(e) Conclude that δ_0 is inadmissible. (3 points)

4. (30 points) Predictive Bayes rule and admissibility.

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\theta, \sigma^2)$ with $\sigma^2 > 0$ known and $\theta \in \mathbb{R}$. Take the prior $\theta \sim N(\mu, \tau^2)$ with $\mu \in \mathbb{R}$ and $\tau^2 > 0$. We wish to predict a future observation $X_{n+1} \sim N(\theta, \sigma^2)$, drawn independently of $X_1, \dots, X_n$ given θ . Use the prediction loss $L^*(z, a) = (z - a)^2$. Let $\bar{X} = (1/n) \sum_{i=1}^n X_i$.

(a) Reduce the prediction problem to a parameter-estimation problem by deriving the induced loss

$$L(\theta, a) = \mathbb{E}_\theta[L^*(X_{n+1}, a)]$$

in closed form. (6 points)

(b) Derive the posterior of θ given $\bar{X}$ as $N(\mu_n, \tau_n^2)$, with explicit formulas for μ_n and τ_n^2 . (8 points)

(c) Derive the Bayes rule $\delta_\pi(\bar{X})$ for the prediction problem under the loss L from (a). Compare it to the Bayes rule for estimating θ under squared-error loss. (6 points)

(d) Show that δ_π is admissible.

(i) Show that δ_π is the unique Bayes rule under L . (4 points)

(ii) Compute $r(\pi, \delta_\pi)$ in closed form and verify it is finite. (6 points)

Best of luck.